Day 1, Semiconductor Training

9:20 Sambutan

- Host

- Dwi Suranto, this program is for meningkatkan kedaulatan chip (tidak bergantung)

  dengan cara membangun talenta Indonesia bersama ARM

  target of 15.000 in 5 years.

  Rantai : design → foundry ~~alean~~ → ATP

  ↳ less costly

  future

  pesan : 1. jangan malu bertanya

  2. disiplin training online

  3. kesempatan jejaring

- Heri Kuswanto

  untuk meningkatkan, mulai dari fundasi, talentanya

- Rifqi Ansari

  kita bisa belajar dari Taiwan, awalnya itu mengirimkan talentanya ke luar negeri, hingga bisa muncul TSMC

- Rektor ITS

  Menerapkan ITS berdampak dan mendunia, dan ITS sendiri sudah berusaha untuk membuat clean room manufacturing 80 M.

10:10

Introduction to ARM, Matt Rushton

- ~~welco~~ SoI the seren, welcoming us
- introdubtion to ARM

ARM → founded in Cambridge late 80's
- ↳ awalnya Acorn Computing (CPU-based company)
- ↳ jadi advanced RISC Machine
- ↳ awalnya 12 engineer
- ↳ now 20 offices

- Bagaima jadi besar?
  - > saat CPU saturasi, cari peluang lain, ~~menunggu~~ keunggulan lain cpu yang low-power (power efficiency), portable, high-performance,

- Sekarang marketnya jadi luas
  - > edge AI
  - > physical AI
  - > cloud AI
- AI Infrastruktur udah dipakai AWS, Nvidia
- mobile phone
- display glasses
- autonomous driving techology

? produk ARM
- arsitektur ARM (the manuals)
- IP (CPU, GPU, system) yang bisa dipakai langsung

- compute subsystem (CSS) gabungan dari IP IP yang bisa dipakai langsung
- silicon : (CPU AGI)

Introduction to CPU ARM Cortex-A

- Architecture : CPU harus apa ketika software berjalan, ISA, exception, memory rule
- micro-architecture : menentukan bagaimana implementasi memenuhi aturan arsitektur
  - > pipeline
  - > memory system
  - > PPA
- ARM
  - > A - profile, highest performance, full OS
  - > R , real-time , bare metal or RTOS
  - > M → compact, energy efficient, bare metal or RTOS
- development

  from 1990 through 2011

  V4 through V8

  > V4, thumb (16-bit)   V8, 64 bit

  V5, DSP   V9,

  V6, memory

  V7, Advanced SIMD

Arm V_4 & V_5

> V_4 : Thumb (16-bit)

> V_5 · DSP improvement, Jazelle java, VFP

V_6 & V_7

> V_6 : memory, multiprocessing core

> V_7 : advanced SIMD, VFP V_3, V_4, Large physical
adress extension

ARM V_8 2011

> 64-bit, and backward compabiliy to 32-bit

> vector extension : SVE dan SVE2

> virtualization, exception, timer, secure EL2

> bfloat : f

V_9, 2021

- CCA (confidential compute architecture), isolated, harware protection
- Machine learning,
  > Matmult

  > SME

  > CME
- SVE, Vect

compact code → SIMD & multicore → 64-bit → security & ML acceleration execution states , AArch 6

- register width
- instruction set
- programming model (memory & exceptions)

A32 & T32 bisa exchanmix, tapi tidak 32bit dan kode 64-bit

exception levels

AArch64: EL0-EL3 (diantaranya, EL3 tertinggi)

AArch32: PE mode

EL3	↑	secure monitor
EL2		hypervisor
EL1	highest	OS
EL0		Applications

how to move between levels

- triggered by core
- can never be taken to EL0
- returned to the same or lower levels

Types

- Synchrous
- IRQ
- FIQ
- SError

AArch PE modes

User , bisa diakses dari kode

FIQ

IRQ

Supervisor

Monitor

Abort

Hyp

Undefined

System

extension x itu minor feature

Biasanya update annualy

Pointer authentication

- otentikasi isi register sebelum digunakan di indirect branch or load

Branch Target Identification

- memory page dilindungi dari eksekusi yang bukan target sebuah branch

Memory Tagging Extension

- deteksi error memory untuk membantu debugging software

Big.little

- mengombinasikan high performance dan low intensity
- tapi keduanya masih bisa berkomunikasi

Cache Coherent Interface

- hardware coherency butuh bus coherence

Dynamiq itu yang implementasi dari ARM menggabungkan banyak big dan LITTLE jadi satu cluster

ARM system IP

- generic interrupt controller, memilih processor yang tepat
- system memory management unit
- AMBA itu arsitektur komunikasi,

Polylion showcase

Edge AI use case

- voice : recognition, processing
- vision : detection, classification
- sensing
- generative AI

Challenge

- power and cooling
- limited RAM
- shared cpu budget
- continuous video
- network connectivity

Gap

faster inference → ethos

-

uses of AI

- inspection product line
- monitoring restricted-area
- review incident

segmentation & pose

Speech [illegible]

- disaster response
- education support
- public service accessibility

Conformer ASR

(1) capture spoken audio
(2) convert to audio features
(3) conformer inference
(4) decode to text

Manfaat Cortex-R

- fast
- real-time
- reliable

Embedded Application

- automotive
- modem
- storage
- healthcare
- industrial

Arsitektur ARMv7R

Mikroarsitektur (implementasinya) : Cortex R4, R5, R7, R8

Out-of-order execution

- misalnya ada urutan instruksi yang datang
- eksekusinya boleh beda urutan

Real-time virtualization

-

64-bit dibutuhkan untuk address memory lebih dari 4GB

Periferal ARM ada beberapa jenis

LLPP : paling ketat 4 cycles max

SPP : setengah ketat 10_s cycles max

Main master : tidak ketat

Memory Hierarchy

TCM , paling ketat 1 cycle

LLRAM , medium 10s of cycles

Main master to Dram , 100s of cycle

microarchitecture

cache is very importand and needed to go get easily [illegible] difference between cortex A & R

- dari POV cache
- dari periferal

Virtualization

- Virtual memory system architecture, agar

Functional safety

- penting.
- diatur standar, misalnya ISO 26262
- dua jenis
  > systematic capability
  > diagnostic "

di ARM ada

- multi level memory protection
- dual core lockstep
- ECC protection.
- bus protection

Lingkungan itu bisa mengubah bit

Arm punya safety package (dokumen)

- manual
- verification report
- failure mode analysis
- diagnostic mode analysis

- development interface report

IEC 61508

Cortex-R

- real time · determinism
- mencakup Arm 8-R and armv7-R
- TCM, LLPP, low latency interrupt
- virtualization; MPU and MMU
- Functional safety

Cortex-M

- paing highest shipping
- armv6 → ArmV8 → trust-zone, co-pro interface

↓ ↓ ↓

armv7 ArmV8.1M memisahkan data / softw yang sudah dipercaya dan belum dipercaya

↓

DSP / FPU

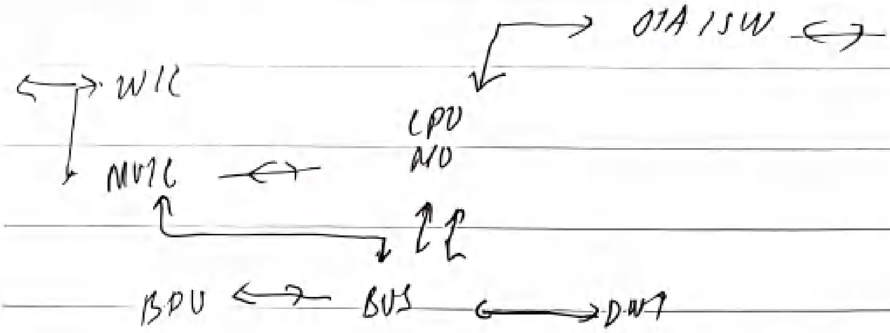
dibuat karen energ efficiency

Cara memilih Cortex-M yang cocok

- cost / die area
- power budget
- DSP
- security
- performance -M2

High performance	M7	M85
performance eff	M3 M4	M33 M55
Lowest power & area	Cortex M0 , M23	

M0



transisi trust zone

· di A, pakai software, secure monitor

· di M, pakai hardware / processor

Trust zone itu seharusnya gak cpu aja, tapi

· memory eksternal juga

· software juga

Helium vector extension

· vector processing engine, multi vektor tapi satu instruksi

· caranya pipeline.

Cara memilih cortex M, dilihat dari

- volume of data

  vibration → voice → vision

- processing rate

Cortex M tools

Corstone

- reference design cortex M55

ARM Mali

Mobile GPU usage

- 3D gaming on mobile devices :
- 2D gaming :
- user interface :
- multimedia : handle high resolution video
- computer vision :
- machine learning : can handle very well because parallel, but a more dedicated accelerator is more
-

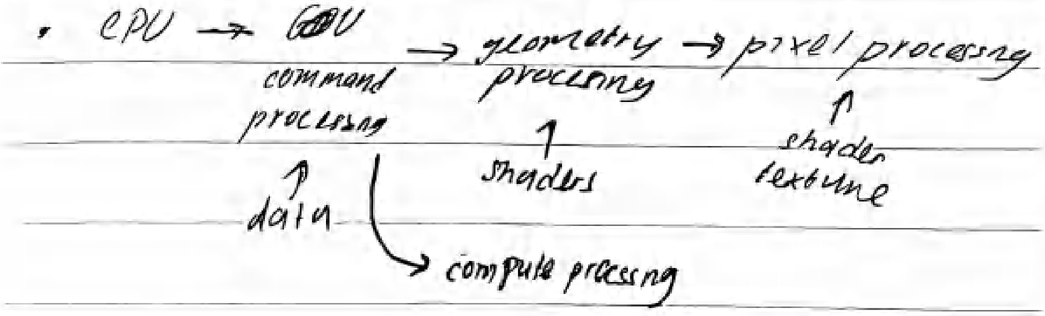
Mobile system

- Cpu $\underset{\text{commands}}{\overset{\text{resources}}{\rightleftarrows}}$ GPU $\xrightarrow{\text{window surface}}$ DPU
- gaming desktop bedanya itu daya yang 500-600 W with that much power, it needs a lot of fan

- pd mobile, power budget biasanya 8 – 15 w
- mobile have a unified memory pool that is shared by all of the components.
- Dram acceses is expensive in power
- power budget harus dibagi ke beberapa komponen, GPU, CPU, DRAM
- power scales super- linearly due to voltage scaling

Rendering Pipeline

- ada mesh → animate → rasterize → color → textures


CPU → GPU
command processing
→ geometry processing → pixel processing
data
shaders
shader texture
compute processing

- geometry processing

vertex shader → tessellation shaders → geometry shader → primitive assembly & primitive culling

object space vertices in

~~• vertex shader~~

- Pixel processing

rasterizing → early ZB test → fragment shader → late ZB test → blend

rasterizer → early ZB test → hidden vertex removal → fragment shader late ZB test → blend

Pipelines and Parallelism

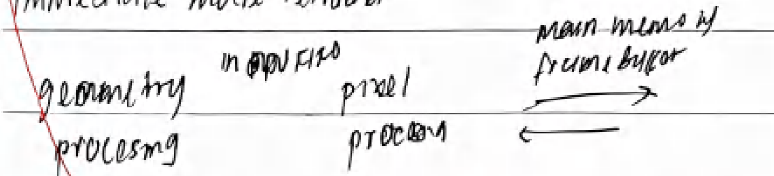
app software	d1	d2			CPU
geometry		d1	d2		GPU
pixel			d1	d2	

- pipeline serialization open GL ES

- pipeline itu ada bottleneck

GPU architecture

Immediate mode renderer


geometry processing
in GPU FIFO
pixel process
main memory framebuffer

less expensive more expensive

Immediate mode pipelining

Tile base rendered, yang biasa dipakai di mobile

main memory tile list

in GPU tile buffer

keuntungan TBR

- Mali shader core
  grafik

- CPU
  > low thread count
  > sensitive to memory latency

SIMD (Single instruction multiple data)

- untuk satu instruksi bisa beberapa operasi data

GPU ciri

- performa lebih diukur dari hasil satu frame ASAP with low power

SIMT (Single program multiple thread)

- setiap instruksi ini dipanggil warp

Keluarga Mali GPU

Utgard → midgard → bifrost → valhall →

- mobile gpu ~~not only~~ powers for everything visual
- power efficiency is the ultime constraint
- TBR = efficiency
- parallel pipelined sustain performance
- shader cores favour throughput over speed
- Mali evolution focuses on smarter design

ARM

- 100% of the connected population
- >350 B chips shipped
- >22 M software developers on ARM
- 50 thousand company collaboration

Future of Edge AI

- edge itu komputasi terjadi di tempat data dibuat dan user berada

randie

AI is scaling

- Trillion of tokens
- 2,4 billion, active gen AI user
- 71,3% AI queries bisa dhandel model lokal

Training → inference
↳ 25-35%
↳ 00-05%

dari tahun ke tahun untuk performa AI sama, memory footprintnya berkurang tahun per tahun

ada Alif Ensemble E8 Devkit, platform AI, arm-based

Executor

- sebelum deply, Pytorch
- setelah deploy

    executorch → delegate → optimized ops → CPU, NPU, GPU

edge constraint

- more inference
- more contextual data
- always on
- limited power
- realtime execution
- finite resources